Aalto University
School of Electrical
Engineering

Master's Programme in Electronics and Electrical Engineering

# Encoding–Surrogate Pairing Effects in Machine Learning-Guided Protein Fitness Optimisation

## A Systematic Evaluation on Sparse and Dense Fitness Landscapes

**Aleksander Heino**

Master's Thesis
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**Abstract**

Machine learning-assisted directed evolution (MLDE) allows researchers to navigate high-dimensional protein fitness landscapes more efficiently than classical random mutagenesis by training a surrogate model on a small labelled set and using it to prioritise the next round of experiments. A critical but unresolved design question is how to pair the sequence representation (encoding) with the surrogate model — and whether the optimal pairing depends on the statistical structure of the fitness landscape being optimised.

This thesis benchmarks four encoding strategies against three surrogate models on two datasets with contrasting landscape structure: the four-site GB1 combinatorial library and the full-protein avGFP dataset from Sarkisyan et al. (2016). We find that ESM2 mean-pooled embeddings provide no significant predictive advantage over sparse one-hot mutation features when paired with tree-based surrogates, and we quantify practical top-variant recovery using Precision@K alongside the standard Spearman rank correlation metric. The results offer actionable guidance for protein engineers choosing ML strategies under realistic laboratory constraints, and raise the question of whether encoding–surrogate recommendations from dense combinatorial benchmarks transfer to sparser, full-protein landscapes.

**Keywords** protein engineering, machine learning, Bayesian optimisation, protein language models, fitness landscapes

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Koneoppimisavusteinen ohjattu evoluutio mahdollistaa proteiinien fitness-maisemien navigoimisen rajoitetulla kokeellisella datalla. Tässä työssä vertaillaan neliää koodausstrategiaa kolmeen surrogaattimalliin kahdella rakenteeltaan erilaisella datasetillä: nelipaikkainen GB1-kombinatorinen kirjasto sekä Sarkisyanin (2016) avGFP-datasetti. Tulokset osoittavat, että ESM2-kielimallien mean-pool-upotukset eivät tuota merkittävää etua yksinkertaisempiin koodausmenetelmiin verrattuna satunnaismetsäsurrogaateilla. Precision@K-metriikka paljastaa merkittävästi alhaisemman käytännön huippuvarianttien löytämiskyvyn kuin Spearman-korrelaatio yksinään antaisi olettaa.

**Avainsanat** proteiinisuunnittelu, koneoppiminen, bayesilainen optimointi, proteiinikielen mallit, fitness-maisema

# Preface

This thesis was carried out at Aalto University during 2025–2026. I would like to thank my supervisor and advisor for their guidance throughout this project, and the broader protein engineering and machine learning communities for making the datasets and tools used here publicly available.

Otaniemi, March 15, 2026
Aleksander Heino

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# Symbols and abbreviations

## Abbreviations

| | |
|---|---|
| AAIndex | amino acid index physicochemical parameter database |
| BO | Bayesian optimisation |
| DKL | deep kernel learning |
| DMS | deep mutational scanning |
| DNN | deep neural network |
| EI | expected improvement |
| ESM | evolutionary scale modelling (protein language model family) |
| FLIP | fitness landscape inference for proteins (benchmark) |
| GP | Gaussian process |
| LM | language model |
| MLDE | machine learning-assisted directed evolution |
| P@K | Precision at K |
| PLM | protein language model |
| RF | random forest |
| RBF | radial basis function (kernel) |
| UCB | upper confidence bound |
| WT | wild type |

# 1 Introduction

Proteins are the molecular machines of life. Whether a protein fluoresces, catalyses a chemical reaction, or binds a target depends exquisitely on its amino acid sequence, yet only a tiny fraction of all possible sequences folds into a functional structure. Directed evolution [**Romero2009**, **Packer2015**] mimics natural selection in the laboratory: sequences are mutated, expressed, assayed for a target property, and the best-performing variants are propagated into the next generation of experiments. While powerful, the approach is inherently combinatorial — the number of possible sequences grows as $20^L$ for a protein of length $L$, making exhaustive experimental screening impossible for all but the shortest peptides.

Machine learning-assisted directed evolution (MLDE) [**Wu2019**] was developed as a response to this combinatorial explosion. Instead of assaying every possible variant, a small labelled set is used to train a surrogate fitness model, which predicts fitness across the unseen sequence space and guides the selection of the next experimental batch. This closed-loop paradigm — sometimes framed as Bayesian optimisation (BO) over sequences — has been validated on several benchmark datasets and shown to identify high-fitness variants with substantially fewer experimental measurements than random screening [**Wu2019**, **Wittmann2021**, **Yang2025**].

A central design choice in any MLDE system is how to represent protein sequences to the surrogate model. Representations range from simple one-hot encodings of mutated positions to dense embeddings from protein language models (PLMs) pre-trained on hundreds of millions of evolutionary sequences [**Lin2023ESM2**]. A recent large-scale benchmark — Active Learning-assisted Directed Evolution (ALDE) [**Yang2025**] — established that this choice is not neutral: the optimal encoding depends on which surrogate model is used. High-dimensional PLM embeddings benefit from deep learning surrogates; simpler surrogates such as random forests or Gaussian processes perform best with lower-dimensional one-hot or physicochemical encodings.

However, ALDE and the majority of MLDE benchmarks were conducted on dense combinatorial libraries — datasets in which every combination of mutations at a small number of fixed sites (typically four) has been measured, yielding on the order of 150,000 variants. Whether these encoding–surrogate pairing recommendations transfer to sparse, full-protein landscapes — where a smaller number of variants carry mutations distributed across the entire sequence — is an open question. The statistical properties of these two landscape types differ qualitatively in ways that may influence which representation is most informative: in a sparse full-protein landscape, mean-pooling PLM embeddings over hundreds of positions dilutes the signal from the few mutated sites, potentially neutralising any advantage of the richer representation.

This thesis investigates that question systematically. We benchmark four encoding strategies and three surrogate models on two datasets with contrasting landscape structure: the four-site GB1 combinatorial library [**Wu2019**] and the full-protein avGFP dataset from Sarkisyan et al. [**Sarkisyan2016**], which contains approximately 52,000 variants spanning 1 to 9 mutations across a 238-residue sequence. In addition to standard metrics (Spearman rank correlation $\rho$, $R^2$), we evaluate Precision@K

— the fraction of the K highest-predicted variants that genuinely belong to the top-K by measured fitness — as an operationally meaningful metric aligned with the experimental selection bottleneck.

**Contributions.** The main contributions of this thesis are:

- a systematic two-dataset benchmark of encoding–surrogate combinations under matched conditions, spanning both dense combinatorial and sparse full-protein landscape regimes;
- empirical evidence that ESM2 mean-pooled embeddings provide no significant advantage over one-hot mutation features when paired with a random forest surrogate, on both landscape types;
- a characterisation of Precision@K as a metric that reveals substantially lower practical top-variant recovery than Spearman $\rho$ implies, and diverges from $\rho$ across encoding–surrogate combinations; and
- an evaluation of delta embeddings (mutant minus wild-type embedding at mutation positions) as a compact, biologically motivated alternative to mean pooling.

**Thesis structure.** Section 2 reviews the relevant literature. Section 3 describes the datasets, encodings, surrogates, and evaluation protocol. Section 4 presents and analyses the experimental results. Section 5 summarises findings, discusses limitations, and outlines future directions.

# 2 Literature Review

## 2.1 From Random Mutagenesis to Machine-Assisted Directed Evolution

The engineering of proteins for non-natural functions has a history stretching back to the early 1990s, when Frances Arnold demonstrated that iterative rounds of random mutagenesis and functional selection could transform the specificity and stability of industrial enzymes without any knowledge of the underlying molecular mechanism [**Romero2009**]. This paradigm — *directed evolution* — exploited natural selection as an optimisation algorithm: diversity is introduced into a gene pool by error-prone PCR, DNA shuffling, or saturation mutagenesis [**Stemmer1994**], and the best-performing variants are identified by a functional assay and carried forward as parents for the next round. The approach proved remarkably general, and its development was recognised with the Nobel Prize in Chemistry in 2018.

Classical directed evolution is limited by experimental throughput. A single round of screening can characterise hundreds to a few thousand variants, but the sequence space grows combinatorially: for a protein of length $L$ with $k$ variable positions, there are up to $20^k$ possible sequences. For the experimentally tractable case of four variable sites, this already yields 160,000 sequences — at the edge of what a well-equipped laboratory can screen in a single campaign. For full-protein engineering with mutations distributed across the entire sequence, exhaustive coverage is impossible.

The introduction of deep mutational scanning (DMS) [**Fowler2014**] partially addressed this bottleneck by coupling high-throughput DNA synthesis and next-generation sequencing to functional selection. In a DMS experiment, a library encoding many thousands of protein variants is simultaneously assayed for function; sequencing before and after selection quantifies the fitness effect of each variant in a single experiment. The Sarkisyan et al. (2016) study [**Sarkisyan2016**] exemplified this approach for the green fluorescent protein (GFP), characterising approximately 52,000 single-protein mutants for fluorescence, providing one of the first quantitative maps of a full-protein fitness landscape. The GB1 dataset from Wu et al. (2019) [**Wu2019**] extended this to combinatorial libraries, measuring all $20^4$ combinations at four coevolutionary positions of the protein G B1 domain.

The integration of machine learning into directed evolution — MLDE [**Wu2019**] — emerged from the observation that DMS data are rich enough to train a surrogate fitness predictor, which can then be queried over unevaluated sequences to prioritise the next experimental batch. Wu et al. demonstrated that a random forest regressor trained on a fraction of GB1 variants could identify top-fitness sequences far more efficiently than random sampling. Subsequent work formalised the MLDE loop as a Bayesian optimisation problem, in which the surrogate provides both a fitness estimate and an uncertainty measure, and an acquisition function trades off exploitation of high-predicted-fitness regions against exploration of uncertain ones [**Romero2009**, **Yang2025**]. More recently, the loop has been extended with active learning strategies that jointly optimise the training set composition and the acquisition policy [**Yang2025**],

and with generative models that propose novel sequences outside the training library rather than selecting from a pre-measured pool [**Stanton2022**].

## 2.2 The Machine Learning Toolkit: Encodings, Surrogates, and Acquisition Functions

Three components define an MLDE system: the representation of each sequence (encoding), the surrogate model that maps representations to predicted fitness, and the acquisition function that selects which sequences to assay next.

**Sequence encodings.** The simplest representation is a *one-hot* encoding of the amino acid identity at each variable position, producing a sparse binary vector. For combinatorial libraries with $k$ variable sites this gives a $20k$-dimensional vector; for full-protein variants, a sparse vector of mutation identities indexed to the full sequence. One-hot encodings contain no learned biological information and treat all amino acid substitutions as equally distant.

Physicochemical encodings such as AAIndex [**Kawashima2000**] or the Georgiev reduced alphabet [**Georgiev2009**] map each amino acid to a small number of empirically measured properties (volume, hydrophobicity, charge, secondary structure propensity), offering a low-dimensional representation with biological inductive bias.

Protein language models (PLMs) represent the most expressive encoding family currently available. Trained by masked-token self-supervision on hundreds of millions of evolutionary protein sequences, models such as ESM1b, ESM2 [**Lin2023ESM2**], and ESM3 [**Hayes2025ESM3**] learn contextualised, position-specific residue representations that implicitly encode structural and functional constraints accumulated over evolutionary time. The ESM2 model family ranges from 8 million to 15 billion parameters; the 650 million parameter variant (ESM2-650M) produces 1280-dimensional per-position embeddings. The simplest approach to deriving a fixed-length sequence representation is to average (*mean-pool*) the per-position embeddings over the full sequence, yielding a 1280-dimensional vector regardless of sequence length. An alternative is to extract only the embeddings at variable positions (*site-specific* extraction), concatenating them to give a higher-dimensional but mutation-focused vector; this is the approach used in the ALDE benchmark [**Yang2025**].

A third possibility, not yet systematically benchmarked in the literature, is the *delta embedding*: computing the difference between the mutant and wild-type embeddings at each mutated position. This approach is motivated by the observation that mean pooling over a full-length protein dilutes the mutation signal across hundreds of invariant positions, potentially explaining why PLM embeddings offer little advantage over one-hot features when paired with simple surrogate models [**Yang2025**].

**Surrogate models.** The three dominant surrogate families in the MLDE literature are random forests (RF) [**Wu2019**], Gaussian processes (GP) [**Romero2009**], and deep neural network (DNN) ensembles [**Lakshminarayanan2017**]. Random forests offer simplicity, robustness to small training sets, and out-of-bag uncertainty estimates via

variance across trees, but their uncertainty estimates are known to be poorly calibrated in sparse input regions. Gaussian processes provide a principled probabilistic framework with closed-form posterior uncertainty, and their uncertainty decays correctly in well-explored regions; however, exact GP inference scales as $O(n^3)$ in the number of training points, limiting their use to datasets of a few thousand samples without approximations. DNN ensembles — typically five or more independently initialised multi-layer perceptrons — offer the expressiveness needed to exploit high-dimensional PLM embeddings and produce uncertainty estimates via ensemble disagreement [**Lakshminarayanan2017**]; they generally require more training data than RF or GP.

Deep kernel learning (DKL) [**Yang2025**] combines a DNN feature extractor with a GP covariance function, aiming to inherit the representational power of deep networks and the calibrated uncertainty of GPs. The ALDE benchmark found DKL with ESM2 site-specific embeddings to be one of the top-performing combinations on GB1 and TrpB, though at substantially higher computational cost than RF or GP.

**Encoding-surrogate pairing effects.** A central finding from Yang et al. (2025) [**Yang2025**] is that the benefit of a richer encoding depends on which surrogate is used. High-dimensional PLM embeddings improve performance when paired with DNN surrogates or DKL, but offer little or no advantage over one-hot or physicochemical encodings when paired with RF or GP. This *encoding-surrogate pairing* effect was also noted earlier by Wittmann et al. [**Wittmann2021**], who found that one-hot encoding outperformed complex representations for non-deep models on GB1. The practical implication is that investing in expensive PLM featurisation is only justified when a correspondingly expressive surrogate is used.

**Acquisition functions.** Given a surrogate that outputs a predicted mean $\hat{\mu}$ and uncertainty $\hat{\sigma}$, an acquisition function decides which sequences to label next. Greedy selection (rank by $\hat{\mu}$, select top-K) ignores uncertainty but is competitive in regimes with large training sets. Upper confidence bound (UCB; select by $\hat{\mu} + \beta\hat{\sigma}$) [**Srinivas2010**] provides a tunable exploration-exploitation trade-off. Thompson sampling draws a fitness function from the posterior and greedily optimises it [**Yang2025**], providing implicit diversity in the selected batch. Expected improvement (EI) maximises the expected gain over the current best observed value.

A non-trivial non-Bayesian baseline is AdaLead [**Sinai2020**], a greedy hill-climbing algorithm with recombination that does not require a calibrated uncertainty model. AdaLead has been shown to match or outperform full Bayesian approaches on several benchmarks, raising the question of when principled uncertainty quantification is worth its additional complexity.

**Zero-shot strategies.** An orthogonal class of approaches requires no labelled fitness data at all: PLM log-likelihood ratios between mutant and wild-type sequences are used directly as fitness proxies [**Notin2022Tranception**, **Hie2024**]. These zero-shot scores are competitive with supervised surrogates in very-low-data regimes

and provide a strong baseline that any supervised method should exceed before the additional cost of labelled data is considered justified. The ProteinGym benchmark [**Notin2023ProteinGym**] provides a comprehensive evaluation of zero-shot predictors across 217 proteins and over 1.5 million variants.

**Two paradigms: generative search versus pool-based selection.** A conceptual divide runs through the protein ML optimisation literature that is rarely stated explicitly but is critical for interpreting results and assessing prior work.

The *generative search* paradigm — exemplified by CbAS [**Brookes2019CbAS**], DbAS [**Brookes2018DbAS**], DynaPPO [**Angermueller2020DynaPPO**], PEX [**Ren2022PEX**], and GGS [**Kirjner2023GGS**] — proposes *novel sequences not present in any measured dataset*. A neural network or PLM is trained on available experimental data and then used as an oracle to score candidate sequences generated by a search algorithm (VAE, RL policy, Gibbs sampler, greedy hill-climbing). The research question is which search algorithm most efficiently finds sequences that the oracle scores as high-fitness. Surrogate quality is treated as a fixed implementation detail, not a variable. Importantly, because the oracle is queried on sequences it may never have seen during training, these methods rely on the oracle's ability to extrapolate — an assumption that is rarely validated against experimental ground truth.

The *pool-based MLDE* paradigm — exemplified by Wu et al. [**Wu2019**], Wittmann et al. [**Wittmann2021**], ALDE [**Yang2025**], and BOES [**Soldat2025BOES**] — selects the best variants from a *fixed pool of already-measured sequences*. A small subset of the pool (the training budget, typically $n$ = 96–5000) is used to train a surrogate; the surrogate ranks the remaining pool members; the top-ranked variants are selected. Ground truth is always a real experimental measurement, not a neural network approximation. The research question is which encoding and surrogate combination best recovers the highest-fitness variants from the pool given the available training budget.

A third category — *low-N generative design* [**Biswas2021LowN**] — combines limited training labels ($n$ = 24–96) with generative search over novel sequences and experimental wet-lab validation of top candidates. This is the closest in spirit to real-world directed evolution campaigns, but it does not directly address the encoding-surrogate pairing question for pool-based prediction.

This thesis belongs to the pool-based MLDE paradigm. Generative search papers frequently use the avGFP dataset but do so by training an oracle on all available measurements and proposing novel sequences — a different task that does not evaluate surrogate quality under a limited labelling budget. Therefore, the extensive use of avGFP in the generative literature does not constitute prior work addressing the research question investigated here.

## 2.3 Benchmark Datasets and Fitness Landscape Structure

Progress in MLDE has been anchored by a small number of landmark datasets. Understanding their structural properties — not just their size — is essential for

interpreting benchmark results and for assessing whether findings generalise across settings.

**GB1 (Wu et al., 2019).** The GB1 dataset [**Wu2019**] is the de facto standard benchmark for combinatorial MLDE. It covers all $20^4$ = 160,000 amino acid combinations at four coevolving positions (V39, D40, G41, V54) in the B1 domain of streptococcal protein G, of which 149,361 variants were successfully measured for binding to IgG-Fc by deep mutational scanning. The landscape is highly epistatic: the fitness effect of any single mutation depends strongly on the identities of the other three positions. Approximately 92% of measured variants have near-zero fitness (the "hole" problem; Wittmann et al. 2021 [**Wittmann2021**]), making the training distribution heavily skewed and posing a challenge for supervised models trained on random subsets.

**TrpB (Johnston et al., 2024).** The TrpB dataset [**Johnston2024**] covers four sites in the indole-3-glycerol phosphate synthase subunit TrpB, providing a second dense combinatorial benchmark with a different epistatic structure from GB1. Johnston et al. characterised approximately 160,000 four-site variants for thermostability and activity, enabling the ALDE benchmark [**Yang2025**] to test whether GB1 findings generalise across combinatorial landscapes.

**avGFP (Sarkisyan et al., 2016).** The avGFP dataset [**Sarkisyan2016**] represents a fundamentally different landscape structure from the combinatorial benchmarks above. Rather than exhaustively measuring all combinations at a few fixed sites, Sarkisyan et al. characterised approximately 52,000 random mutants of the Aequorea victoria green fluorescent protein (avGFP), each carrying between one and nine mutations distributed across the full 238-residue sequence. The resulting landscape is *sparse* — it covers only a tiny fraction of the sequence space reachable by a small number of mutations — and heterogeneous in mutation count and location.

The avGFP dataset has been used in both paradigms described above. In the search paradigm, a CNN or similar model is trained on all 52,000 measurements and used as an oracle; the research question is which search algorithm finds sequences the oracle predicts as highly fluorescent. In the surrogate paradigm relevant to this thesis, only a small subset of measurements is available as training data, and the question is which encoding-surrogate combination best ranks the remaining variants. The latter has received substantially less attention despite being the operationally relevant setting for wet-lab protein engineering.

Several structural features distinguish avGFP from combinatorial benchmarks. First, the training distribution is not a random subset of a combinatorial grid but a biased sample of an enormous space; models must generalise to sequences with different mutation counts and positions simultaneously. Second, the fraction of near-zero-fitness variants (hole density) is lower than in GB1, partially reducing the class imbalance problem. Third, mean-pooling PLM embeddings over the full 238-residue sequence dilutes the mutation signal across more than 230 invariant positions, a structural artefact that may explain why PLM featurisation offers limited

gains over one-hot features when the surrogate is a random forest (see Section 4.1). Fourth, ALDE's site-specific embedding — which extracts embeddings at a fixed small set of variable positions — does not directly apply to avGFP, where variable positions differ from variant to variant.

**ProteinGym and FLIP.** The ProteinGym benchmark [**Notin2023ProteinGym**] aggregates DMS data for 217 proteins and evaluates zero-shot fitness predictors. The FLIP benchmark [**Dallago2021FLIP**] provides standardised splits for supervised fitness prediction. Both benchmarks are valuable for broad coverage, but neither addresses the encoding-surrogate pairing question directly, and neither disentangles the dense-combinatorial versus sparse-full-protein landscape regimes.

**Evaluation metrics.** Spearman's rank correlation coefficient $\rho$ is the dominant evaluation metric in both zero-shot benchmarking [**Notin2023ProteinGym**] and MLDE studies [**Wu2019**, **Yang2025**]. It measures the monotonic association between predicted and measured fitness ranks across the full test set. The $R^2$ coefficient of determination is also widely reported.

However, in the context of active sequence selection, what matters most operationally is not global rank correlation but the quality of the top of the predicted distribution: a model may achieve high $\rho$ overall yet fail to concentrate the genuinely best variants at the top of its ranked list. Precision@K — the fraction of the K highest-predicted variants that actually belong to the top-K by measured fitness — captures this directly and aligns naturally with the experimental bottleneck of a fixed synthesis-and-assay budget. Despite this operational relevance, Precision@K is underreported in the published MLDE literature; most papers cite Spearman $\rho$ or $R^2$ as their primary metrics.

## 2.4 Current State of the Art and Open Questions

The ALDE benchmark [**Yang2025**] is the most systematic published comparison of encoding-surrogate-acquisition function combinations to date. Testing all combinations of four encodings (one-hot, AAIndex/Georgiev, ESM2 site-specific), four surrogate types (boosting ensemble, GP, DNN ensemble, DKL), and two acquisition functions (greedy, Thompson sampling) on GB1 and TrpB, it established that: (i) the DNN ensemble with Thompson sampling is the best overall strategy across conditions; (ii) ESM2 site-specific embeddings improve performance when paired with DNN or DKL surrogates; and (iii) non-deep surrogates benefit little or not at all from richer encodings. These findings are consistent with the earlier observation by Wittmann et al. [**Wittmann2021**] that one-hot encoding outperforms more complex representations for non-deep models on GB1.

EVOLVEpro [**Liu2024EVOLVEpro**] demonstrated a practical active learning system using ESM2-15B mean-pooled embeddings with a random forest surrogate, achieving measurable fitness improvements across five diverse protein targets in an

iterative wet-lab setting. EVOLVEpro also reported a notable failure mode: PLM zero-shot fitness scores (log-likelihood ratios) showed near-zero or negative correlation with experimentally measured activity for several proteins, highlighting that evolutionary fitness and specific biological function are distinct quantities.

ESM3 [**Hayes2025ESM3**] introduced a multimodal generative PLM jointly modelling sequence, structure, and function tokens. At 98 billion parameters, the full model is computationally prohibitive for most research groups; however, smaller ESM C variants (300 million parameters) are more accessible and may provide embeddings competitive with ESM2 at lower computational cost. Whether ESM3 embeddings provide material benefits over ESM2 in supervised fitness prediction relevant to MLDE is not yet established.

Three important questions remain unresolved at the time of writing. First, essentially all systematic encoding-surrogate benchmarks have been conducted on dense combinatorial libraries (GB1, TrpB); whether optimal pairings transfer to sparse full-protein landscapes such as avGFP is unknown. Second, the delta embedding approach has been proposed as a biologically motivated alternative to mean pooling but has not been benchmarked. Third, Precision@K as a primary evaluation metric for active learning in protein engineering remains underused; it is not clear how strongly encoding-surrogate combinations differ on this metric versus Spearman $\rho$.

## 2.5 Research Question and Contributions

The preceding review motivates the central research question of this thesis:

> *Do the encoding-surrogate pairing recommendations derived from dense combinatorial fitness landscapes (GB1) transfer to sparse, full-protein landscapes (avGFP), and how does the choice of encoding-surrogate pair affect Precision@K as an operationally meaningful evaluation metric?*

More concretely, we test whether the relative ranking of four encoding strategies (one-hot mutation features, ESM2 mean-pooled, ESM2 delta, AAIndex physicochemical) across three surrogate models (random forest, Gaussian process, DNN ensemble) is consistent between GB1 and avGFP, and whether Spearman $\rho$ and Precision@K give concordant or divergent rankings.

The work makes four concrete contributions. First, it provides the first systematic encoding-surrogate benchmark that spans both the dense combinatorial and sparse full-protein landscape regimes. Second, it evaluates Precision@K alongside Spearman $\rho$ and $R^2$, quantifying the gap between global correlation and practical top-variant recovery. Third, it tests the delta embedding hypothesis: that subtracting wild-type embeddings at mutation positions recovers signal lost by mean pooling, potentially restoring a PLM advantage in the sparse regime. Fourth, it characterises learning curves for all combinations, providing guidance on the minimum labelling budget at which each encoding-surrogate pair is competitive.

# 3 Methods

## 3.1 Datasets

**GB1.** The GB1 dataset is obtained from the SaProtHub repository (SaProtHub/Dataset-GB1-fitness) and the FLIP benchmark [**Dallago2021FLIP**], using the standard two-versus-rest split in which any variant containing at least one of the four wild-type amino acids is held out for testing.

**avGFP.** The avGFP dataset is obtained from the MaveDB repository using the Sarkisyan et al. accession (MaveDB:00000055-a-1) and is preprocessed to retain only variants with valid fluorescence measurements and between one and nine amino acid substitutions relative to the wild type.

## 3.2 Sequence Encodings

**One-hot mutation features.** Each variant is represented by a sparse binary vector of length proportional to the number of variable positions, encoding the identity of the amino acid change at each mutated site. For GB1 this yields a $4 \times 20 = 80$-dimensional vector; for avGFP the vector length varies with mutation count and is zero-padded to a fixed maximum.

**ESM2 mean-pooled embeddings.** Each variant sequence is passed through ESM2-650M (Meta AI; Lin et al. 2023 [**Lin2023ESM2**]) and the per-position embeddings are averaged over all residues to produce a 1280-dimensional sequence representation. Embeddings are precomputed and stored; ESM2 is not fine-tuned.

**ESM2 delta embeddings.** The mutant embedding is computed by subtracting the wild-type embedding at each mutated position: $\delta_i = h_i^{\text{mut}} - h_i^{\text{WT}}$ for each mutated position $i$. The per-position deltas are averaged to produce a 1280-dimensional summary vector. This approach is designed to concentrate the representation on mutation-induced changes and reduce the dilution effect of averaging over invariant positions.

**AAIndex physicochemical features.** Each amino acid substitution is encoded using a selected subset of AAIndex [**Kawashima2000**] indices covering hydrophobicity, volume, charge, and secondary structure propensity.

## 3.3 Surrogate Models

**Random forest.** A random forest regressor is trained using scikit-learn with $n_{\text{estimators}} = 100$ and default hyperparameters.

**Gaussian process.** An exact GP with a radial basis function (RBF) kernel is trained using GPyTorch [**Gardner2018**]. Kernel hyperparameters are optimised by marginal likelihood maximisation. Inputs are standardised to zero mean and unit variance before fitting.

**DNN ensemble.** An ensemble of five independent multi-layer perceptrons is trained with identical architecture but different random initialisations [**Lakshminarayanan2017**]. Each network has two hidden layers of 256 units with ReLU activations. Uncertainty is estimated as the standard deviation of ensemble predictions.

## 3.4 Evaluation Protocol

All models are evaluated in an *offline* Bayesian optimisation setting: a fixed training set of size $n$ is sampled uniformly at random from the full dataset, and the surrogate is used to rank the remaining (unseen) variants. Each configuration is repeated across five independent random seeds.

Three metrics are reported: Spearman rank correlation $\rho$ between predicted and measured fitness on the held-out set; $R^2$ on the same set; and Precision@K for $K \in \{10, 100, 1000\}$. Precision@K is defined as

$$\mathrm{P@}K = \frac{|\{k : \mathrm{rank}_{\mathrm{pred}}(k) \leq K\} \cap \{k : \mathrm{rank}_{\mathrm{true}}(k) \leq K\}|}{K}, \tag{1}$$

where $\mathrm{rank}_{\mathrm{pred}}(k)$ is the rank of variant $k$ in the predicted ordering and $\mathrm{rank}_{\mathrm{true}}(k)$ is its rank by measured fitness.

Learning curves are generated by varying training set size $n \in \{50, 100, 200, 500, 1000, 2000, 5000, 1000$ with five seeds per configuration.

# 4 Results

## 4.1 ESM2 Mean Pooling versus One-Hot Features on avGFP

**Preliminary finding (avGFP, RF, n=10,000).** A random forest surrogate trained on ESM2 mean-pooled embeddings (1280-dimensional, ESM2-650M) achieves $R^2 = 0.469 \pm 0.003$ on the avGFP held-out set, virtually identical to the same surrogate trained on one-hot mutation features ($R^2 = 0.466 \pm 0.001$, $n = 5$ independent runs). The difference is not statistically significant. Despite comparable $R^2$, Precision@100 is low for both encoding strategies (3–9%), indicating that the models correctly rank the broad fitness distribution but struggle to concentrate the top-100 highest-fitness variants at the top of their predicted lists.

## 4.2 Learning Curves

## 4.3 GB1 Benchmark

## 4.4 Delta Embedding Evaluation

# 5 Discussion and Conclusions

## 5.1 Summary of Findings

## 5.2 Implications for Protein Engineering Practice

## 5.3 Limitations

## 5.4 Future Work

Promising directions include: replacing the random forest with an AdaLead greedy baseline to test whether BO-style acquisition adds value over simple hill-climbing; extending the benchmark to the TrpB dataset to confirm GB1 findings generalise across dense combinatorial landscapes; evaluating ESM3 embeddings as replacements for ESM2; and simulating multi-round active learning with observation noise to approximate the online setting studied in the wet-lab MLDE literature.

# A Experimental Configuration